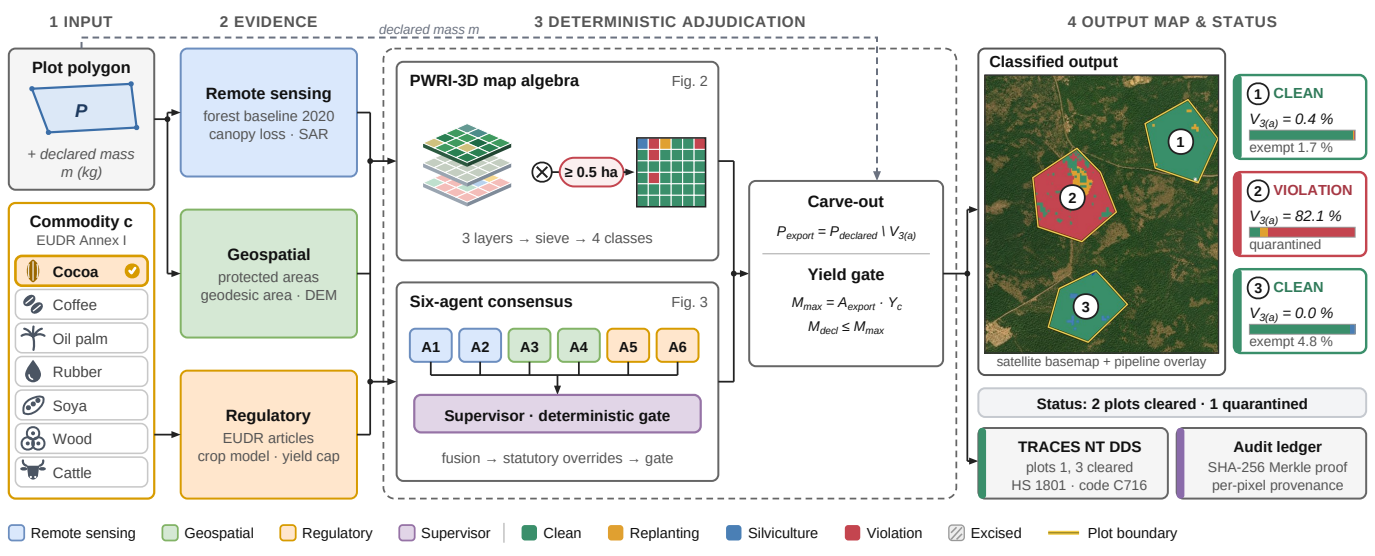

1 INPUT
2 EVIDENCE
declared mass m
3 DETERMINISTIC ADJUDICATION
4 OUTPUT MAP & STATUS
Plot polygon
P
+ declared mass m (kg)
Commodity c
EUDR Annex I
Cocoa
Coffee
Oil palm
Rubber
Soya
Wood
Cattle
Remote sensing
forest baseline 2020
canopy loss · SAR
Geospatial
protected areas
geodesic area · DEM
Regulatory
EUDR articles
crop model · yield cap
PWRI-3D map algebra
Fig. 2
≥ 0.5 ha
3 layers → sieve → 4 classes
Six-agent consensus
Fig. 3
A1
A2
A3
A4
A5
A6
Supervisor · deterministic gate
fusion → statutory overrides → gate
Carve-out
$P_{export} = P_{declared} \setminus V_{3(a)}$
Yield gate
$M_{max} = A_{export} \cdot Y_c$
$M_{decl} \le M_{max}$
Classified output
1
2
3
satellite basemap + pipeline overlay
1 CLEAN
$V_{3(a)} = 0.4$ %
exempt 1.7 %
2 VIOLATION
$V_{3(a)} = 82.1$ %
quarantined
3 CLEAN
$V_{3(a)} = 0.0$ %
exempt 4.8 %
Status: 2 plots cleared · 1 quarantined
TRACES NT DDS
plots 1, 3 cleared
HS 1801 · code C716
Audit ledger
SHA-256 Merkle proof
per-pixel provenance
Remote sensing
Geospatial
Regulatory
Supervisor
Clean
Replanting
Silviculture
Violation
Excised
Plot boundary